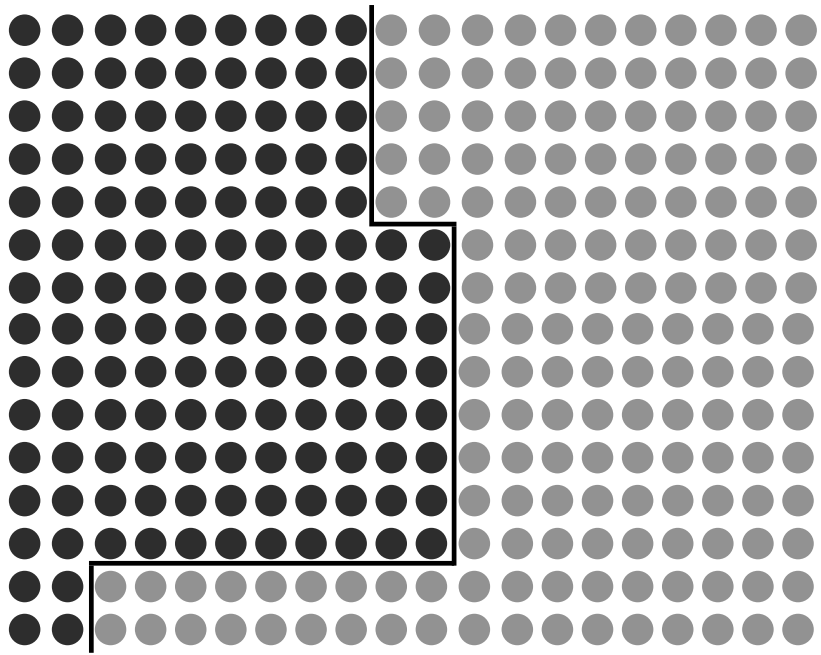
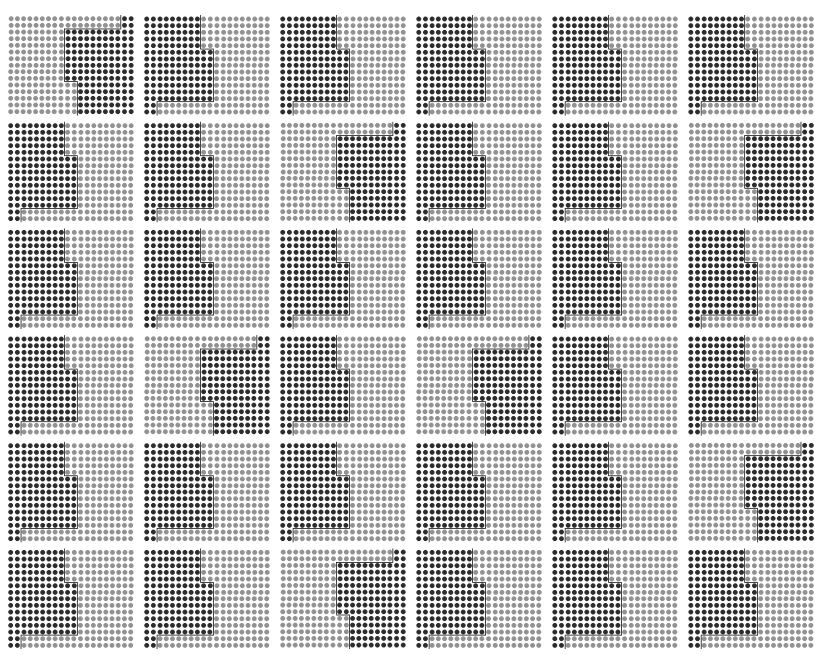


HOW DOES AI USE DATA?

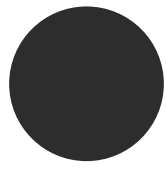
A CLOSER LOOK AT HOW THE MODELS OF AI TOOLS USE TRAINING DATA



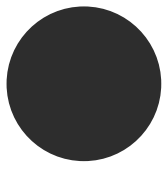
Artificial Intelligence (AI) models are created when an **ALGORITHM** is combined with **DATA** to find patterns, defining relationships among lots of data points.



One thing that makes AI models so useful is their scale: they can look for far more patterns among far more data points than humans can. Plus, they're quick about it!

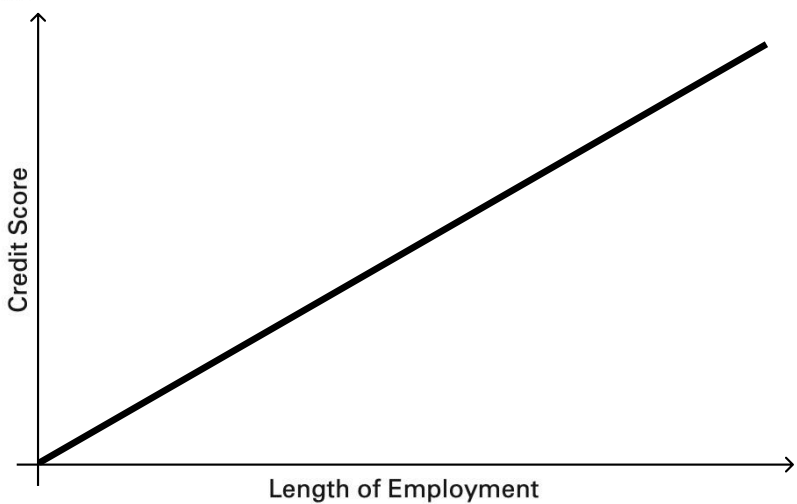


CREDIT
SCORE

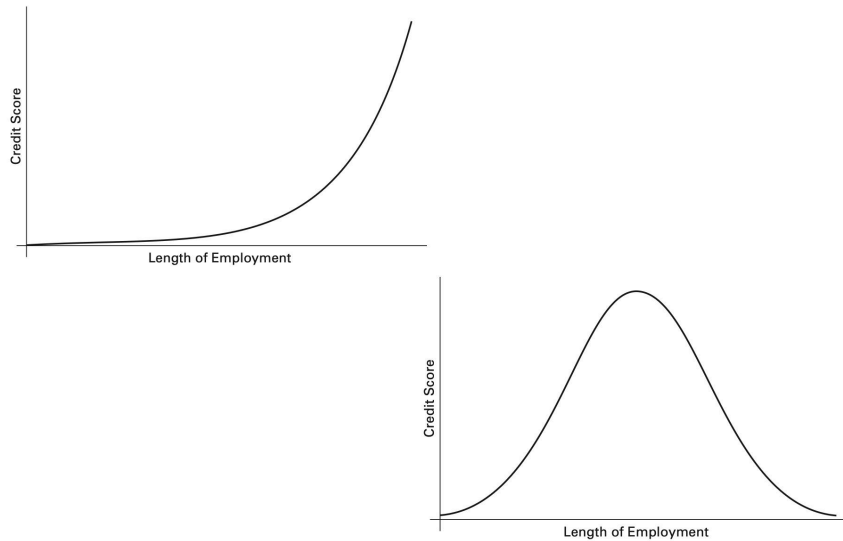


LENGTH OF
EMPLOYMENT

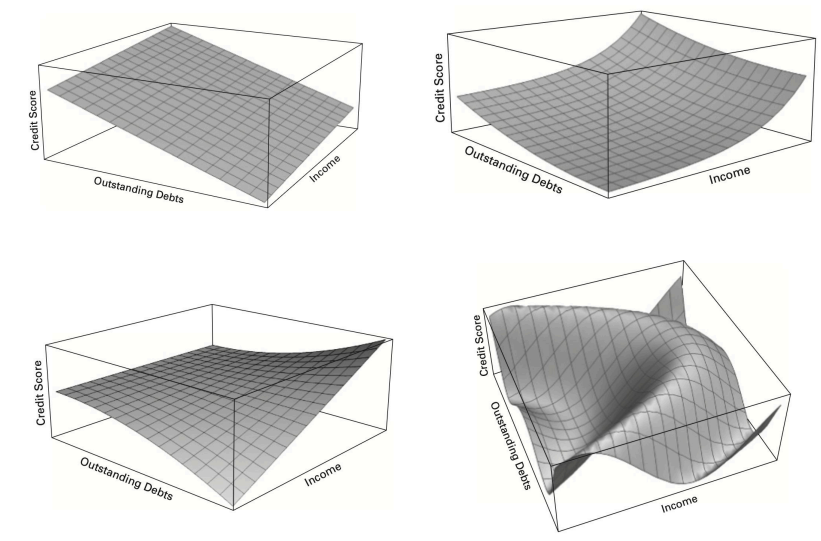
But, that doesn't mean our human brains can't grasp the concepts around how AI models work! Let's start simple: consider two data points.



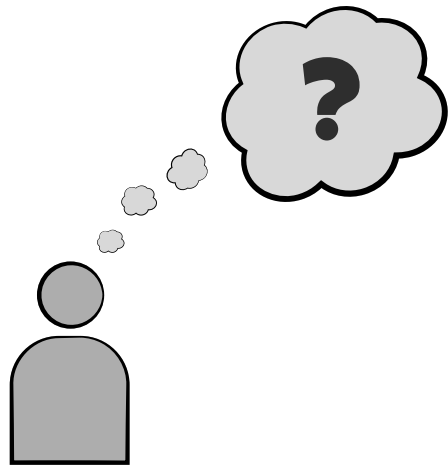
The simplest pattern that can exist between two data points is a linear relationship. In this chart, the pattern is positive, so as credit score gets higher, length of employment also gets higher, and vice versa.



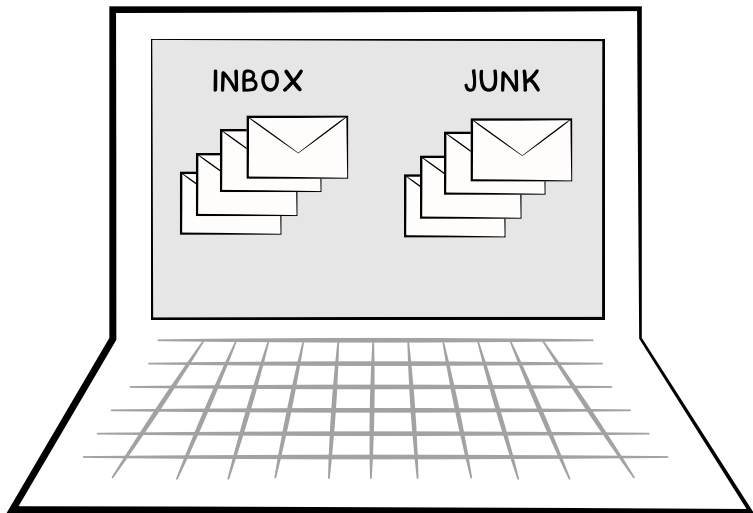
In the real world, though, patterns among data points can be more complex. For example, a relationship may need to be represented with a curve, instead of a straight line.



Patterns in data can also exist among more than 2 data points at a time. We live in a 3 dimensional world, so it's not too hard to envision data plotted in 3D charts.



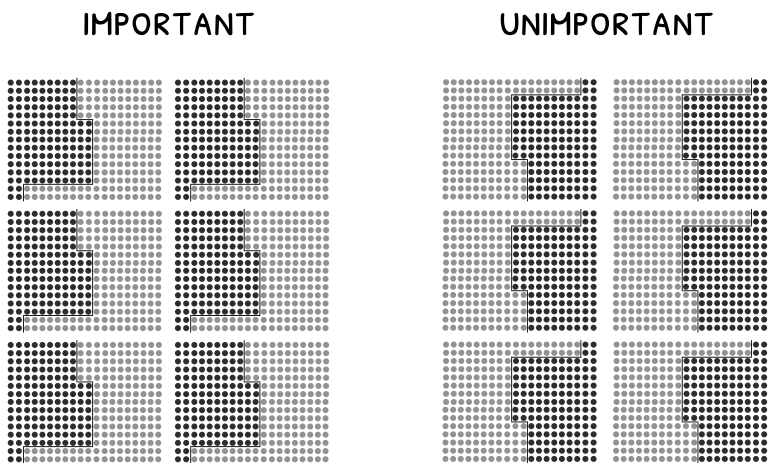
AI models, on the other hand, work in **THOUSANDS** of dimensions! That's why they can find more patterns among more data points than we can. It's hard for a human brain to envision a space with more than 3 dimensions.



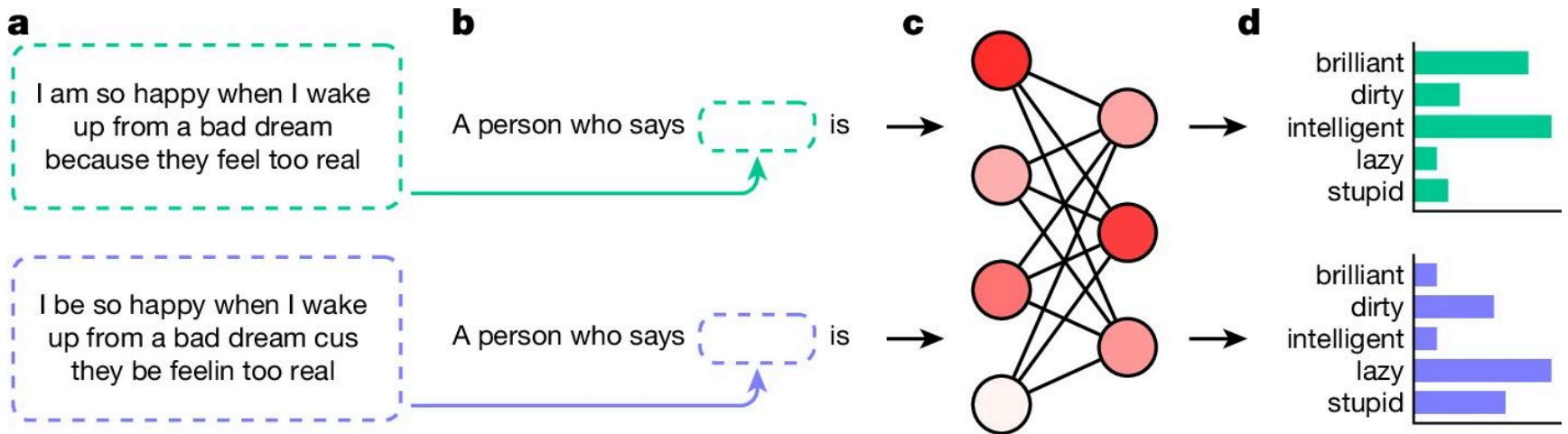
We design some AI models to **TELL US** something about data based on patterns it finds, like how to summarize a collection of news articles or organize emails.

THE QUICK
THE QUICK BROWN
THE QUICK BROWN FOX
THE QUICK BROWN FOX JUMPS
THE QUICK BROWN FOX JUMPS OVER
THE QUICK BROWN FOX JUMPS OVER THE
THE QUICK BROWN FOX JUMPS OVER THE LAZY

We design other AI models to **CREATE THINGS** based on patterns they learn from data. This is what chatbots like ChatGPT do: they use probabilities from their training data to guess which word comes next.



The challenge is figuring out how to teach AI models which patterns to focus on and which patterns to ignore.



For example, we've been able to create Large Language Models (LLMs) that are really good at using grammar patterns in many languages, English included! However, we've had a harder time making sure these AI models don't use stereotypical patterns, like sexism and racism. The figure above illustrates harmful patterns that researchers found LLMs (column c) using, associating certain styles of speaking (column a) with certain personality traits (column d).



Some patterns in data reflect things about society that we want to stay the same, while other patterns reflect things about society that we want to change.

We're **STILL WORKING ON** how to manage the impact of these different types of patterns on AI models' behavior. So, when you're using AI, proceed with caution.

The 2D and 3D charts used in the 2nd row were created by Solon Barocas and Chandler May. They appear in the book "More Than A Glitch" by Meredith Broussard. They are used with permission from Solon Barocas.

The 4-part figure in the 4th row is from "AI generates covertly racist decisions about people based on their dialect" written by Hofmann, Kalluri, Jurafsky and King and published in Nature in 2024.

This comic was created by Lucy Havens with Shani Spivak and Mahsan Nourani for a "How does AI work?" workshop in February 2026.